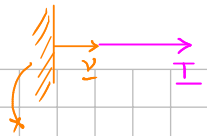


$$(\sigma_{ij}) = \begin{pmatrix} 1 & 0 & -1 \\ 0 & -1 & 0 \\ -1 & 0 & 1 \end{pmatrix}$$

Fórmula de Cauchy $\underline{\underline{\tau}}^{(i)} = \underline{\underline{\tau}} \underline{\underline{v}}^{(i)}$

$$\underline{\underline{\tau}}^{(i)} = \lambda^{(i)} \underline{\underline{v}}^{(i)}$$

$$\left\{ \begin{array}{l} \underline{\underline{\tau}} \underline{\underline{v}}^{(i)} = \lambda \underline{\underline{v}}^{(i)} \\ (\underline{\underline{\tau}} // \underline{\underline{v}}^{(i)}) \end{array} \right.$$


$$\underline{\underline{\tau}} \underline{\underline{v}}^{(i)} - \lambda^{(i)} \underline{\underline{v}}^{(i)} = 0$$

$$\underline{\underline{\tau}} \underline{\underline{v}}^{(i)} - \lambda^{(i)} \underline{\underline{I}} \underline{\underline{v}}^{(i)} = 0$$

$$(\underline{\underline{\tau}} - \lambda^{(i)} \underline{\underline{I}}) \underline{\underline{v}}^{(i)} = 0$$

$\underline{\underline{A}} \Rightarrow \det(\underline{\underline{A}}) = 0 \rightarrow$ Ecuación característica

$\hookrightarrow$ singular para que no resulte solo en la solución trivial de $\underline{\underline{v}}^{(i)}$

$$\tau_{kj}(\underline{\underline{v}}^{(i)})_j - \lambda^{(i)} \underbrace{(\underline{\underline{v}}^{(i)})_k}_{\frac{\partial \tau_{kj}(\underline{\underline{v}}^{(i)})}{\partial \lambda^{(i)}}} = 0$$

$$|\tau_{ij} - \sigma \delta_{ij}| = \begin{vmatrix} \tau_{11} - \sigma & \tau_{12} & \tau_{13} \\ \tau_{21} & \tau_{22} - \sigma & \tau_{23} \\ \tau_{31} & \tau_{32} & \tau_{33} - \sigma \end{vmatrix} = -\sigma^3 + I_1 \sigma^2 - I_2 \sigma + I_3 = 0,$$

$$I_1 = \tau_{11} + \tau_{22} + \tau_{33},$$

$$I_2 = \begin{vmatrix} \tau_{22} & \tau_{23} \\ \tau_{32} & \tau_{33} \end{vmatrix} + \begin{vmatrix} \tau_{33} & \tau_{31} \\ \tau_{13} & \tau_{11} \end{vmatrix} + \begin{vmatrix} \tau_{11} & \tau_{12} \\ \tau_{21} & \tau_{22} \end{vmatrix},$$

$$I_3 = \begin{vmatrix} \tau_{11} & \tau_{12} & \tau_{13} \\ \tau_{21} & \tau_{22} & \tau_{23} \\ \tau_{31} & \tau_{32} & \tau_{33} \end{vmatrix}.$$

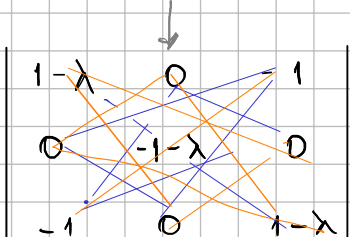
$$\rightarrow I_1(\underline{\underline{\tau}}) = \text{tr}(\underline{\underline{\tau}})$$

$$I_2(\underline{\underline{\tau}}) = \frac{1}{2} [\text{tr}^2(\underline{\underline{\tau}}) - \text{tr}(\underline{\underline{\tau}}^2)]$$

$$\rightarrow I_3(\underline{\underline{\tau}}) = \det(\underline{\underline{\tau}})$$

$\rightarrow$ traza $\rightarrow$ suma de los componentes de la diagonal de $\underline{\underline{\tau}}$

$$\underline{\underline{\tau}} \cdot \underline{\underline{\tau}}$$



$$= (1-\lambda) \cdot (-1-\lambda) \cdot (1-\lambda) - (-1-\lambda)$$

$$= \lambda (-\lambda^2 + \lambda + 2) \rightarrow \lambda = 0 = \lambda_2$$

$$\lambda_1 > \lambda_2 > \lambda_3$$

$$\rightarrow \text{Cuadrático} \rightarrow \lambda = -1 = \lambda_3$$

$$\rightarrow \lambda = 2 = \lambda_1$$

$$\det(\underline{\underline{A}}) = (-1)^{2+2} \cdot (-1-\lambda) \cdot \begin{vmatrix} 1-\lambda & -1 \\ -1 & 1-\lambda \end{vmatrix} = -(1+\lambda) \cdot [(1-\lambda)^2 - 1]$$

$$\hookrightarrow \lambda = -1$$

$$\hookrightarrow \lambda^{(2)} = -1$$

$$\rightarrow (1-\lambda)^2 - 1 = 0$$

$$1 - 2\lambda + \lambda^2 - 1 = 0$$

$$\lambda(-2+\lambda) = 0$$

$$\hookrightarrow \lambda = 0 \rightarrow \lambda = 2$$

$$\hookrightarrow \lambda^{(2)} = 0 \quad \hookrightarrow \lambda^{(1)} = 1$$

$$(I - \lambda^{(i)} A) \underline{v}^{(i)} = \underline{0}$$

$$\|\underline{v}^{(i)}\| = \sqrt{v_1^{(i)^2} + v_2^{(i)^2} + v_3^{(i)^2}} = 1$$

$$\begin{bmatrix} 1 - \lambda^{(i)} & 0 & -1 \\ 0 & -1 - \lambda^{(i)} & 0 \\ -1 & 0 & 1 - \lambda^{(i)} \end{bmatrix} \cdot \begin{bmatrix} v_1^{(i)} \\ v_2^{(i)} \\ v_3^{(i)} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{cases} (1 - \lambda^{(i)}) \cdot v_1^{(i)} - v_3^{(i)} = 0 \\ (-1 - \lambda^{(i)}) v_2^{(i)} = 0 \\ -1 v_1^{(i)} + (1 - \lambda^{(i)}) v_3^{(i)} = 0 \end{cases} \Rightarrow \lambda^{(i)} \neq -1 \Rightarrow v_2^{(i)} = 0$$

$$\bullet \lambda^{(1)} = 2 \Rightarrow \begin{cases} v_2^{(1)} = 0 \\ -v_1^{(1)} - v_3^{(1)} = 0 \\ -v_1^{(1)} - v_3^{(1)} = 0 \end{cases} \Rightarrow v_1^{(1)} = -v_3^{(1)}$$

$$\|\underline{v}^{(1)}\| = \sqrt{(-v_3^{(1)})^2 + 0^2 + (v_3^{(1)})^2} = 1 \Rightarrow \sqrt{2} |v_3^{(1)}| = 1 \Rightarrow v_3^{(1)} = \pm 1/\sqrt{2}$$

$$\Downarrow$$

$$v_1^{(1)} = \mp 1/\sqrt{2}$$

$$\bullet \lambda^{(2)} = 0 \Rightarrow \begin{cases} v_2^{(2)} = 0 \\ v_1^{(2)} - v_3^{(2)} = 0 \\ -v_1^{(2)} + v_3^{(2)} = 0 \end{cases} \Rightarrow v_1^{(2)} = v_3^{(2)}$$

$$\|\underline{v}^{(2)}\| = 1 \Rightarrow v_3^{(2)} = \pm \frac{1}{\sqrt{2}} \Rightarrow v_1^{(2)} = \pm \frac{1}{\sqrt{2}}$$

$$\bullet \lambda^{(3)} = -1 \Rightarrow 2v_1^{(3)} - v_3^{(3)} = 0 \Rightarrow v_3^{(3)} = 2v_1^{(3)}$$

$$0 \cdot v_2^{(3)} = 0 \Rightarrow v_2^{(3)} \in \mathbb{R} \quad (\text{keinen } \mathbb{C} \text{ vorsektor})$$

$$-v_1^{(3)} + 2v_3^{(3)} = 0 \Rightarrow -v_1^{(3)} + 2 \cdot 2v_1^{(3)} = 0 \Rightarrow v_1^{(3)} = 0 \Rightarrow v_3^{(3)} = 0$$

$$\|\underline{v}^{(3)}\| = 1 \Rightarrow |\underline{v}_2^{(3)}| = 1 \Rightarrow \underline{v}_2^{(3)} = \pm 1$$

Autovectores o direcciones principales:

$$\underline{v}^{(1)} = \begin{bmatrix} -\alpha \\ 0 \\ \alpha \end{bmatrix}, \quad \underline{v}^{(2)} = \begin{bmatrix} \alpha \\ 0 \\ \alpha \end{bmatrix}, \quad \underline{v}^{(3)} = \begin{bmatrix} 0 \\ \gamma \\ 0 \end{bmatrix}$$

$\underline{e}_1$ $\underline{e}_2$ $\underline{e}_3$

$$\alpha = \pm \frac{1}{\sqrt{2}}$$

$$\gamma = \pm 1$$

$$\{\alpha\} \rightarrow \{\gamma\}$$

fila → columna

$$P_{ij} = \underline{e}_i \cdot \underline{e}_j = (\underline{e}_i)_j = \begin{bmatrix} (\underline{e}_1)_1 & (\underline{e}_1)_2 & (\underline{e}_1)_3 \\ \underline{e}_2 \\ \underline{e}_3 \end{bmatrix}$$

$$P = \begin{bmatrix} -\alpha & 0 & \alpha \\ \alpha & 0 & \alpha \\ 0 & \gamma & 0 \end{bmatrix} \Rightarrow \det(P) = (-1)^{3+2} \cdot \gamma \cdot \begin{vmatrix} -\alpha & \alpha \\ \alpha & \alpha \end{vmatrix}$$

$$= -\gamma \cdot (-\alpha^2 - \alpha^2) = 2\gamma\alpha^2$$

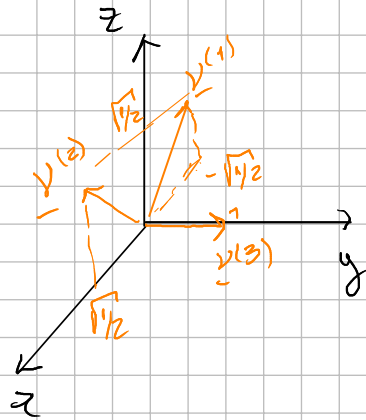
$$\rightarrow \underline{e}_3 = \underline{e}_1 \times \underline{e}_2$$

$$\det(P) = 1 = 2\gamma\alpha^2 \Rightarrow |\alpha| = \frac{1}{\sqrt{2\gamma}} \Rightarrow \gamma \in \mathbb{R} \Rightarrow \gamma = +1$$

destrógras

$$|\alpha| = \frac{1}{\sqrt{2}} \Rightarrow \alpha = \pm \sqrt{\frac{1}{2}}$$

$$\alpha = +\sqrt{\frac{1}{2}} \Rightarrow \underline{v}^{(1)} = \begin{bmatrix} -\sqrt{1/2} \\ 0 \\ \sqrt{1/2} \end{bmatrix}, \quad \underline{v}^{(2)} = \begin{bmatrix} \sqrt{1/2} \\ 0 \\ \sqrt{1/2} \end{bmatrix}, \quad \underline{v}^{(3)} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$



$$\det(\underline{\underline{P}}) = -1 \Rightarrow |\alpha| = \frac{1}{\sqrt{-2\gamma}} \Rightarrow \gamma = -1$$

levógira

$$\Rightarrow \alpha = \pm \sqrt{\frac{1}{2}}$$